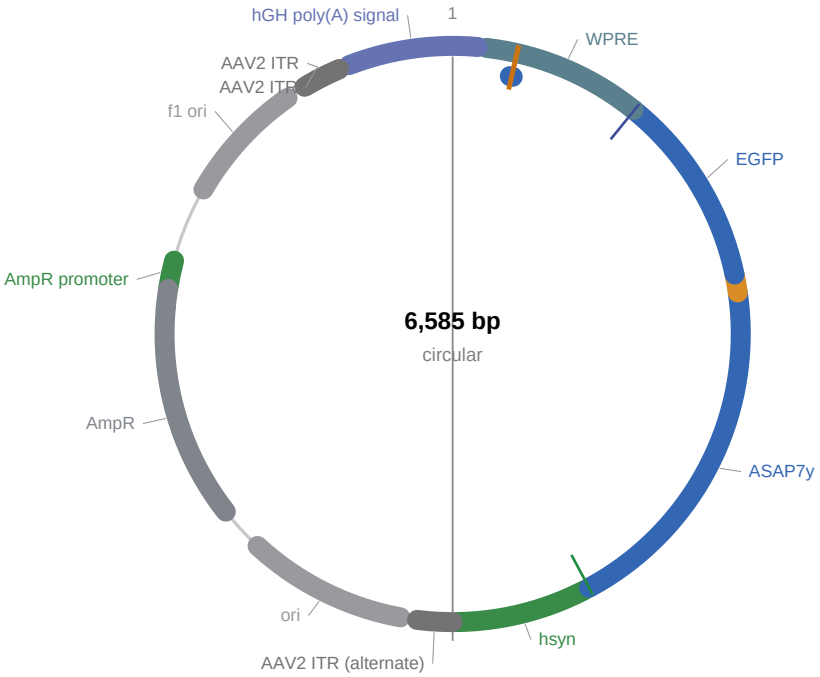
# untitled

PCR + NEBuilder HiFi assembly

| | |
|---|---|
| **Final construct** | 6,585 bp, circular |
| **Backbone** | pBJ027_pAAV_hSyn_ASAP3_ (5,829 bp, Bill Jia) |
| **Architecture** | hsyn - ASAP7y - P2A - EGFP - WPRE - Factor Xa site - hGH poly(A) signal - sprBJ029 (HGH-pA REV) - AAV2 ITR - AAV2 ITR - f1 ori - AmpR promoter - AmpR - AmpR - ori - AAV2 ITR (alternate) |
| **Workload** | 12 steps, 5 PCR(s), 1 assembly reaction(s) |
| **Time** | 4.8 h hands-on, 2.8 days elapsed |
| **Suggested name** | pAAV-hSyn-ASAP7y-P2A-EGFP |

## Designed construct

**untitled** 6,585 bp




Outer ring reads forward, inner ring reverse. Position 1 is at the top.

- no stop codon at 3,052 in Factor Xa site
- no ATG (ATT) at 3,061 in Factor Xa site

## Reagents and consumables

Quantities are per reaction. Check against the current package insert for the kits in use.

| Item | Amount |
|---|---|
| 5x high-fidelity reaction buffer | 10 uL |
| 10 mM dNTPs | 1 uL |
| 10 uM forward primer | 2.5 uL |
| 10 uM reverse primer | 2.5 uL |
| template plasmid (1-10 ng) | 1 uL |
| high-fidelity DNA polymerase | 0.5 uL |
| nuclease-free water | to 50 uL |
| completed PCR reaction | all |
| DpnI | 1 uL |
| agarose gel | 0.8-1 % (1.5 % below 500 bp) |
| DNA stain | per kit |

| | |
|---|---|
| gel extraction kit | per kit |
| PCR or digest reaction | all |
| spin column or SPRI beads | per kit |
| 2x assembly master mix | 10 uL |
| linearised vector | 50-100 ng |
| each insert | 2:1 molar ratio over vector |
| competent cells | 50 uL |
| assembly or ligation product | 2 uL |
| SOC medium | 950 uL |
| colonies | 4-8 |
| screening primer pair | per reaction |
| Taq master mix | per reaction |
| miniprep DNA | per facility spec |
| sequencing primers | one per junction |

## Primers to order

Lower case is the overlap or adapter tail; upper case binds the template. The quoted Tm is for the template-binding region only, which is what sets the annealing temperature.

| Name | Dir | nt | Tm C | GC | Sequence (5' to 3') |
|---|---|---|---|---|---|
| backbone3-F | F | 41 | 59.7 | 28% | cgagctgtacaagtaaAATCAACCTCTGGATTACAAAATTT |
| backbone3-R | R | 18 | 59.8 | 56% | AGGAACCCCTAGTGATGG |
| hsyn-F | F | 48 | 61.3 | 58% | tcactaggggttcctgcggccgcacgcgtGTGTCTAGACTGCAGAGGG |
| hsyn-R | R | 18 | 63.2 | 56% | TTCTCGACTGCGCTCTCA |
| ASAP7y-F | F | 51 | 60.5 | 56% | gagcgcagtcgagaaggtaccggatccgccaccATGGAGACGACTGTGAGG |
| ASAP7y-R | R | 22 | 60.0 | 41% | GGTTACCACTTCAAGTTGTTTC |
| P2A-F | F | 33 | 68.0 | 72% | cttgaagtggtaaccGGCTCCGGAGCCACGAAC |
| P2A-R | R | 18 | 62.5 | 61% | GGGACCGGGGTTTTCTTC |
| EGFP-F | F | 33 | 65.0 | 67% | gaaaaccccggtcccGTGAGCAAGGGCGAGGAG |
| EGFP-R | R | 19 | 58.9 | 47% | TTACTTGTACAGCTCGTCC |

## Protocol

### 1. PCR Amplify the 4,004 bp backbone3 arc (1,955-130) from pBJ027_pAAV_hSyn_ASAP3_ with primers backbone3-F / backbone3-R

annealing 60 C; extension 2:00 for 4,004 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

**PCR reaction (50 uL, high-fidelity polymerase)**

| | |
|---|---|
| 10 uL | 5x high-fidelity reaction buffer |
| 1 uL | 10 mM dNTPs |
| 2.5 uL | 10 uM forward primer |
| 2.5 uL | 10 uM reverse primer |
| 1 uL | template plasmid (1-10 ng) |
| 0.5 uL | high-fidelity DNA polymerase |
| to 50 uL | nuclease-free water |

### 2. PCR Amplify hsyn (485 bp) from pBJ027_pAAV_hSyn_ASAP3_ with primers hsyn-F / hsyn-R

annealing 61 C; extension 0:15 for 485 bp at 30 s/kb; 70% GC, add the high-GC enhancer; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

**PCR reaction (50 uL, high-fidelity polymerase)**

| | |
|---|---|
| 10 uL | 5x high-fidelity reaction buffer |
| 1 uL | 10 mM dNTPs |
| 2.5 uL | 10 uM forward primer |
| 2.5 uL | 10 uM reverse primer |
| 1 uL | template plasmid (1-10 ng) |
| 0.5 uL | high-fidelity DNA polymerase |

| | |
|---|---|
| to 50 uL | nuclease-free water |

### 3. PCR Amplify ASAP7y (1281 bp) from paav-ef1a-dio-asap7y-wpre_ with primers ASAP7y-F / ASAP7y-R

annealing 60 C; extension 0:38 for 1,281 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

**PCR reaction (50 uL, high-fidelity polymerase)**

| | |
|---|---|
| 10 uL | 5x high-fidelity reaction buffer |
| 1 uL | 10 mM dNTPs |
| 2.5 uL | 10 uM forward primer |
| 2.5 uL | 10 uM reverse primer |
| 1 uL | template plasmid (1-10 ng) |
| 0.5 uL | high-fidelity DNA polymerase |
| to 50 uL | nuclease-free water |

### 4. PCR Amplify P2A (66 bp) from pBJ230_pLJM1_JNKKTR_mK with primers P2A-F / P2A-R

annealing 62 C; primer Tm differ by 6 C, consider a gradient; extension 0:15 for 66 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

**PCR reaction (50 uL, high-fidelity polymerase)**

| | |
|---|---|
| 10 uL | 5x high-fidelity reaction buffer |
| 1 uL | 10 mM dNTPs |
| 2.5 uL | 10 uM forward primer |
| 2.5 uL | 10 uM reverse primer |
| 1 uL | template plasmid (1-10 ng) |
| 0.5 uL | high-fidelity DNA polymerase |
| to 50 uL | nuclease-free water |

### 5. PCR Amplify EGFP (717 bp) from GUBD_opsin1 with primers EGFP-F / EGFP-R

annealing 59 C; primer Tm differ by 6 C, consider a gradient; extension 0:22 for 717 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

**PCR reaction (50 uL, high-fidelity polymerase)**

| | |
|---|---|
| 10 uL | 5x high-fidelity reaction buffer |
| 1 uL | 10 mM dNTPs |
| 2.5 uL | 10 uM forward primer |
| 2.5 uL | 10 uM reverse primer |
| 1 uL | template plasmid (1-10 ng) |
| 0.5 uL | high-fidelity DNA polymerase |
| to 50 uL | nuclease-free water |

### 6. DPNI_DIGEST Digest the PCR reactions with DpnI to remove the parental plasmid templates (GUBD_opsin1, pBJ027_pAAV_hSyn_ASAP3_, pBJ230_pLJM1_JNKKTR_mK, paav-ef1a-dio-asap7y-wpre_)

37 C for 30-60 min, added directly to the PCR. Only cuts methylated (dam+) parental plasmid, not the amplicon.

### 7. GEL_EXTRACTION Gel purify hsyn (multiple products or a size-selection requirement)

Run far enough to resolve the bands, excise the correct band on a blue/long-wave transilluminator, elute in 20-30 uL.

### 8. DNA_CLEANUP Column purify the remaining PCR products

Elute in 20-30 uL. Quantify before assembly.

### 9. HI_FI_ASSEMBLY Assemble 5 fragments (hsyn + ASAP7y + P2A + EGFP + backbone3) using NEBuilder HiFi, 25 bp overlaps

50 C for 15 min (2-3 fragments) or 60 min (4 or more). Keep total fragment input near 0.03-0.2 pmol.

**HiFi / Gibson assembly (20 uL)**

| | |
|---|---|
| 10 uL | 2x assembly master mix |
| 50-100 ng | linearised vector |

```
2:1 molar ratio over vector each insert
to 20 uL              nuclease-free water
```

### 10. `TRANSFORMATION` Transform the assembly into competent cells and plate on the appropriate antibiotic

Ice 30 min, 42 C for 30 s, ice 5 min, add SOC, recover 37 C for 1 h with shaking, plate on selective agar, incubate overnight at 37 C.

**Transformation**

```
50 uL                 competent cells
2 uL                  assembly or ligation product
950 uL                SOC medium
```

### 11. `COLONY_SCREEN` Screen 4-8 colonies by colony PCR, or diagnostic digest with NotI

Patch each colony before screening. Confirm candidates by diagnostic digest and set up overnight cultures.

### 12. `SEQUENCE_VERIFY` Sanger sequence across every junction, then confirm the full insert

Sequence across every junction first, then the full insert. Check the translated fusion, not only the DNA.

# Detail

Everything behind the protocol above: how the design was validated, where each fragment comes from, what the other routes would cost, and the final sequence.

## Validation

0 error(s), 7 warning(s), 5 informational.

### Warnings

- dna: hsyn: 11 candidate parts matched; using 'hsyn'
- dna: P2A: 12 candidate parts matched; using 'P2A'
- dna: EGFP: the best-ranked 'EGFP' annotation (part 695, 717 bp) carries no stop codon and nothing translated follows it; used part 557 (717 bp) instead, which ends in one
- dna: EGFP: 12 candidate parts matched; using 'EGFP'
- dna: the backbone feature 'Kozak sequence' straddles an insertion point and was disrupted
- protein: fusion (Factor Xa site) does not begin with ATG
- protein: fusion (Factor Xa site) does not end in a stop codon

### Checks

- protein: predicted fusion protein (ASAP7y + P2A + EGFP): 687 aa
- protein: predicted fusion protein (Factor Xa site): 4 aa
- dna: unique sites available for diagnostic digest: AgeI, ApaI, BamHI, BglII, BsrGI, ClaI, KpnI, MluI, NcoI, XbaI, XhoI
- packaging: carries the full AAV transfer plasmid element set
- verification: read back 4 of 4 parts from the assembled sequence, each once, in the specified order and orientation

## Candidate parents

The request read as an ordered architecture: hsyn - ASAP7y - P2A - EGFP. The plasmid to start from is whichever already carries the longest leading run of it; the rest is covered by as few donors as possible.

### Start from

| Plasmid | Size | Owner | Already has |
|---|---|---|---|
| pBJ027_pAAV_hSyn_ASAP3_ | 5,829 bp | Bill Jia | hsyn (as hSyn promoter), ASAP7y (as Translation 648-1954) |
| paav-hsyn-dio-asap6c-kv-wpre_confirmed | 6,331 bp | Byung Hun Lee | hsyn (as hSyn promoter), ASAP7y (as ASAP4c 413G) |
| YD007-pAAV-hSyn-CaMKAR | 5,486 bp | Yangdong Wang | hsyn (as hSyn promoter) |
| Predicted_YA1406 | 5,539 bp | Yoav Adam | hsyn |
| YA1406_pAAV-hSyn | 5,539 bp | Yoav Adam | hsyn |
| HFS24_c1maxi | 5,599 bp | Felix Sigmund | hsyn (as hSyn promoter) |

### Then take the rest from 1 donor

- P2A - EGFP
- PCR_prdt2_p2A_Ch (1,820 bp, PZ) - has P2A, EGFP (as GFP)
- YQ0104 extraction (3,460 bp, Yitong Qi) - has P2A, EGFP (as GFP)
- YA1401_pENTR_Opt (5,956 bp, Yoav Adam) - has P2A, EGFP (as GFP)
- SLF169 (6,856 bp, Sami Farhi) - has P2A, EGFP (as GFP)

## Cloning simulation

The primers below were amplified off the named templates and the products assembled, to check the steps build this construct rather than merely describe it.

Result: 6,585 bp product, 4/4 requested parts present - identical to the designed sequence

| Part | Length | In the product |
|---|---|---|
| hsyn | 485 bp | yes (100%) |
| ASAP7y | 1,281 bp | yes (100%) |

| | | |
|---|---|---|
| P2A | 66 bp | yes (100%) |
| EGFP | 717 bp | yes (100%) |

## Read-back

The assembled sequence re-annotated from scratch and compared with what was asked for, independently of the bookkeeping that built it.

read back 4 of 4 parts from the assembled sequence, each once, in the specified order and orientation

## Route warnings

- template pBJ027_pAAV_hSyn_ASAP3_ is not sequence-verified
- template paav-ef1a-dio-asap7y-wpre_ is not sequence-verified
- P2A is only 66 bp; rather than a separate PCR, add it to an adjacent fragment's primer tail or order it as annealed oligos
- template pBJ230_pLJM1_JNKKTR_mK is not sequence-verified
- template GUBD_opsin1 is not sequence-verified
- junction 10 (AAV2 ITR|f1 ori): overlap arm is 76% GC; HiFi joining may be poor
- junction 11 (f1 ori|AmpR promoter): overlap arm is only 16% GC; overlap Tm will be low
- junction 12 (AmpR promoter|AmpR): overlap arm is only 24% GC; overlap Tm will be low
- primer backbone3-F: GC content 28% is low
- primer backbone3-F: 6 nt self-complementarity at the 3' end; primer-dimer risk
- primer backbone3-R: annealing region occurs 2 times in the template; expect mispriming
- primer hsyn-F: GC-rich 3' end (4+ of the last 5 bases); may misprime
- primer ASAP7y-F: possible hairpin with a 6 bp stem
- primer P2A-F: annealing Tm 68.0 C is well above the 60 C target
- primer P2A-F: GC content 72% is high
- primer EGFP-F: GC content 67% is high
- primer EGFP-R: GC-rich 3' end (4+ of the last 5 bases); may misprime

## Fragments and sources

| Fragment | Length | Source template | Synthesis |
|---|---|---|---|
| hsyn | 485 bp | pBJ027_pAAV_hSyn_ASAP3_ | no |
| ASAP7y | 1,281 bp | paav-ef1a-dio-asap7y-wpre_ | no |
| P2A | 66 bp | pBJ230_pLJM1_JNKKTR_mK | no |
| EGFP | 717 bp | GUBD_opsin1 | no |

## Junctions

| # | Left | Right | Frame shift | Notes |
|---|---|---|---|---|
| 1 | hsyn | ASAP7y | 0 | ok |
| 2 | ASAP7y | P2A | 0 | ok |
| 3 | P2A | EGFP | 0 | ok |
| 4 | EGFP | WPRE | 0 | ok |
| 5 | WPRE | Factor Xa site | 0 | ok |
| 6 | Factor Xa site | hGH poly(A) signal | 0 | ok |
| 7 | hGH poly(A) signal | sprBJ029 (HGH-pA REV) | 0 | ok |
| 8 | sprBJ029 (HGH-pA REV) | AAV2 ITR | 0 | ok |
| 9 | AAV2 ITR | AAV2 ITR | 0 | ok |
| 10 | AAV2 ITR | f1 ori | 0 | overlap arm is 76% GC; HiFi joining may be poor |
| 11 | f1 ori | AmpR promoter | 0 | overlap arm is only 16% GC; overlap Tm will be low |
| 12 | AmpR promoter | AmpR | 0 | overlap arm is only 24% GC; overlap Tm will be low |
| 13 | AmpR | AmpR | 0 | ok |
| 14 | AmpR | ori | 0 | ok |
| 15 | ori | AAV2 ITR (alternate) | 0 | ok |

## Route comparison

| Route | Steps | PCR | Assembly | Hands-on | Elapsed | Cost | Usable |
|---|---|---|---|---|---|---|---|
| PCR + NEBuilder HiFi assembly (this one) | 12 | 5 | 1 | 4.8 h | 2.8 d | 108 | yes |
| Swap by digest + one PCR + HiFi assembly | 0 | 0 | 0 | 0.0 h | 0.0 d | 1000 | no |
| Swap by PCR + HiFi assembly (2 parents) | 0 | 0 | 0 | 0.0 h | 0.0 d | 1000 | no |
| Swap by restriction digest + ligation (2 parents) | 0 | 0 | 0 | 0.0 h | 0.0 d | 1000 | no |
| Restriction digest + ligation | 0 | 0 | 0 | 0.0 h | 0.0 d | 1000 | no |

Swap by digest + one PCR + HiFi assembly was not usable: 2 of the parts this design needs are absent from pBJ027_pAAV_hSyn_ASAP3_ (P2A, EGFP), and a single swap replaces a single region, so no edit of this plasmid produces the construct asked for; build it from parts, or start from a recipient that already carries the rest

Swap by PCR + HiFi assembly (2 parents) was not usable: 2 of the parts this design needs are absent from pBJ027_pAAV_hSyn_ASAP3_ (P2A, EGFP), and a single swap replaces a single region, so no edit of this plasmid produces the construct asked for; build it from parts, or start from a recipient that already carries the rest

Swap by restriction digest + ligation (2 parents) was not usable: 2 of the parts this design needs are absent from pBJ027_pAAV_hSyn_ASAP3_ (P2A, EGFP), and a single swap replaces a single region, so no edit of this plasmid produces the construct asked for; build it from parts, or start from a recipient that already carries the rest

Restriction digest + ligation was not usable: the insert is made of 4 pieces from 4 donor plasmid(s); a ligation of that many fragments is not practical, so this construct needs the PCR/HiFi route (or a staged build of two pieces at a time)

## Final sequence

6,585 bp, 54.0% GC, circular.

```
1  gcggccgcac gcgtgtgtct agactgcaga gggccctgcg tatgagtgca agtgggtttt
61  aggaccagga tgaggcgggg tgggggtgcc tacctgacga ccgaccccga cccactggac
121  aagcacccaa cccccattcc ccaaattgcg catcccctat cagagagggg gaggggaaac
181  aggatgcggc gaggcgcgtg cgcactgcca gcttcagcac cgcggacagt gccttcgccc
241  ccgcctggcg gcgcgcgcca ccgccgcctc agcactgaag gcgcgctgac gtcactcgcc
301  ggtccccccgc aaactcccct tcccggccac cttggtcgcg tccgcgccgc cgccggccca
361  gccggaccgc accacgcgag gcgcgagata ggggggcacg ggcgcgacca tctgcgctgc
421  ggcgccggcg actcagcgct gcctcagtct gcggtgggca gcggaggagt cgtgtcgtgc
481  ctgagagcgc agtcgagaag gtaccggatc cgccaccatg gagacgactg tgaggtatga
541  acaggggtca gagctcacta aaacttcgag ctctccaaca gcagatgagc ccacgataaa
601  gattgatgat ggtcgtgatg agggtaatga acaagacagc tgttccaata ccattaggag
661  aaaaatttcc ccgtttgtga tgtcatttgg attcagagta tttggagttg tgcttatcat
721  tgtagacatc atagtggtga ttgtggatct ggccatcagt gagaagaaaa gaggcattag
781  agagattctt gaaggtgttt ccctggctat agcactcttc ttccttgttg atgttctcat
841  gagagtgttt gttgaaggct tcaagaacta tttccggtcc aaactgaata ctttggatgc
901  agtcatagta gtgggcactc tgctaattgc tatgacctac tccttctctg ctcttgctgc
961  ctttaaatcc gagaacgtgt atattaccgc ggataaacag aaaaacggca ttaaagcgaa
1021  ctttaccgtg attcataacg tggaagatgg cagcgtgcag ctggcggatc attatcagca
1081  gaacaccccg attggcgatg gcccggtgct gctgccggat aaccattatc tgagctacca
1141  gaccgtgctg agcaaagatc cgaacgaaaa acgcgatcac atggtgctgc tggaagacgt
1201  gaccgcagcg ggcattacac acggcatgga tgaactgtat ggcggcaccg gcggcagcgc
1261  gagccagggc gaagaactgt ttaccggcgt ggtgccgatt ctggtggaac tggatggcga
1321  tgtgaacggc cataaattta gcgtgcgcgg cgaaggcgaa ggcgatgcga ccattggcaa
1381  actgaccctg aaatttattt gcaccaccgg caaactaccg gtgccgtggc cgaccctggt
1441  gaccacctta acctatggcg tgcagtgctt tagccgctat ccggatcata tgaaacgcca
1501  tgatttttt aaaagcgcga tgccggaagg ctatgtgcag gaacgcacca ttagctttaa
1561  agatgatggc aaatataaaa cccgcgcggt ggtgaaattt gaaggcgata ccctggtgaa
1621  ccgcattgaa ctgaaaggca ccgattttaa agaagatggc aacattctgg ggcataaact
1681  ggaatataac attgatcaga tgccgcagat ggttactctt cttcgagttc tgagaattgt
1741  tatcttaata agagtgattc gcctggcttc acagaagaaa caacttgaag tggtaaccgg
1801  ctccggagcc acgaacttct ctctgttaaa gcaagcagga gacgtggaag aaaaccccgg
1861  tcccgtgagc aagggcgagg agctgttcac cggggtggtg cccatcctgg tcgagctgga
1921  cggcgacgta aacggccaca agttcagcgt gtccggcgag ggcgagggcg atgccaccta
1981  cggcaagctg accctgaagt tcatttgcac caccggcaag ctgcccgtgc cctggcccac
2041  cctcgtgacc accctgacct acggcgtgca gtgcttcagc cgctaccccg accacatgaa
2101  gcagcacgac ttcttcaagt ccgccatgcc cgaaggctac gtccaggagc gcaccatctt
2161  cttcaaggac gacggcaact acaagacccg cgccgaggtg aagttcgagg gcgacaccct
2221  ggtgaaccgc atcgagctga agggcatcga cttcaaggag gacggcaaca tcctggggca
2281  caagctggag tacaactaca acagccacaa cgtctatatc atggccgaca agcagaagaa
2341  cggcatcaag gtgaacttca agatccgcca caacatcgag gacggcagcg tgcagctcgc
```

```
2401  cgaccactac cagcagaaca cccccatcgg cgacggcccc gtgctgctgc ccgacaacca
2461  ctacctgagc acccagtccg ccctgagcaa agaccccaac gagaagcgcg atcacatggt
2521  cctgctggag ttcgtgaccg ccgccgggat cactctcggc atggacgagc tgtacaagta
2581  aaatcaacct ctggattaca aaatttgtga aagattgact ggtattctta actatgttgc
2641  tccttttacg ctatgtggat acgctgcttt aatgcctttg tatcatgcta ttgcttcccg
2701  tatggctttc attttctcct ccttgtataa atcctggttg ctgtctcttt atgaggagtt
2761  gtggcccgtt gtcaggcaac gtggcgtggt gtgcactgtg tttgctgacg caacccccac
2821  tggttggggc attgccacca cctgtcagct cctttccggg actttcgctt tccccctccc
2881  tattgccacg gcggaactca tcgccgcctg ccttgcccgc tgctggacag gggctcggct
2941  gttgggcact gacaattccg tggtgttgtc ggggaaatca tcgtcctttc cttggctgct
3001  cgcctatgtt gccacctgga ttctgcgcgg gacgtccttc tgctacgtcc cttcggccct
3061  caatccagcg gaccttcctt cccgcggcct gctgccggct ctgcggcctc ttccgcgtct
3121  tcgccttcgc cctcagacga gtcggatctc cctttgggcc gcctccccgc atcgataccg
3181  agcgctgctc gagagatcta cgggtggcat ccctgtgacc cctccccagt gcctctcctg
3241  gccctggaag ttgccactcc agtgcccacc agccttgtcc taataaaatt aagttgcatc
3301  attttgtctg actaggtgtc cttctataat attatggggt ggaggggggt ggtatggagc
3361  aaggggcaag ttgggaagac aacctgtagg gcctgcgggg tctattggga accaagctgg
3421  agtgcagtgg cacaatcttg gctcactgca atctccgcct cctgggttca agcgattctc
3481  ctgcctcagc ctcccgagtt gttgggattc caggcatgca tgaccaggct cagctaattt
3541  ttgtttttt ggtagagacg gggtttcacc atattggcca ggctggtctc caactcctaa
3601  tctcaggtga tctacccacc ttggcctccc aaattgctgg gattacaggc gtgaaccact
3661  gctcccttcc ctgtccttct gattttgtag gtaaccacgt gcggaccgag cggccgcagg
3721  aacccctagt gatggagttg gccactccct ctctgcgcgc tcgctcgctc actgaggccg
3781  ggcgaccaaa ggtcgcccga cgcccgggct ttgcccgggc ggcctcagtg agcgagcgag
3841  cgcgcagctg cctgcagggg cgcctgatgc ggtattttct ccttacgcat ctgtgcggta
3901  tttcacaccg catacgtcaa agcaaccata gtacgcgccc tgtagcggcg cattaagcgc
3961  ggcgggtgtg gtggttacgc gcagcgtgac cgctacactt gccagcgcct tagcgcccgc
4021  tcctttcgct ttcttccctt cctttctcgc cacgttcgcc ggctttcccc gtcaagctct
4081  aaatcggggg ctccctttag ggttccgatt tagtgcttta cggcacctcg accccaaaaa
4141  acttgatttg ggtgatggtt cacgtagtgg gccatcgccc tgatagacgg tttttcgccc
4201  tttgacgttg gagtccacgt tctttaatag tggactcttg ttccaaactg gaacaacact
4261  caactctatc tcgggctatt cttttgattt ataagggatt ttgccgattt cggtctattg
4321  gttaaaaaat gagctgattt aacaaaaatt taacgcgaat tttaacaaaa tattaacgtt
4381  tacaatttta tggtgcactc tcagtacaat ctgctctgat gccgcatagt taagccagcc
4441  ccgacacccg ccaacacccg ctgacgcgcc ctgacgggct tgtctgctcc cggcatccgc
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4681  tatttgttta ttttttctaaa tacattcaaa tatgtatccg ctcatgagac aataaccctg
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